

2024



Notorious Studios Builds a Scalable Game Analytics Pipeline for Player Research on AWS

Learn how Notorious Studios, an indie video game startup, captures near real-time player insights using a scalable solution on AWS.

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Overview

Playtesting is a critical step in the video game design process, helping developers tune and improve the player experience. [Notorious Studios](#), a game studio, wanted to accelerate the design of its upcoming game, *Legacy: Steel & Sorcery*, with game analytics. “We wanted to get real data on what our players were doing during our playtest, and validate design decisions early in development,” says David Perez, principal game play engineer at Notorious Studios.

To capture near real-time insights relating to player preferences and gameplay optimization, Notorious Studios built a scalable game analytics pipeline on Amazon Web Services (AWS).



Opportunity | Creating a Game Analytics Pipeline for Notorious Studios

Notorious Studios was formed by experienced game developers who have a passion for action-based fantasy games. Some of these developers came from Blizzard and worked on popular games such as *Overwatch* and *World of Warcraft*. *Legacy: Steel & Sorcery*, the company's first title, is an online fantasy-action role-playing game. Players can select characters and classes to play as, with the option to go on solo adventures or play with their friends.

When Notorious Studios began playtests for *Legacy: Steel & Sorcery*, it didn't have a structured solution for tracking player data. To get gameplay insights, its design team would consolidate data manually, which could take up to 3 days. Notorious Studios wanted to combine both qualitative and quantitative data in near real time. “Since we are a small team, we needed a solution that was simple to deploy, integrate, and maintain,” says Perez.

Scalability was also a factor, because the team expects that its playtests will grow as the game is released. So, Notorious Studios used [Guidance for Game Analytics Pipeline on AWS](#), which provides



architecture and sample code that helps game developers implement a codified, modular, and serverless analytics pipeline that ingests telemetry events from game clients and backend services.

Solution | Building a Modular Pipeline Using AWS CDK

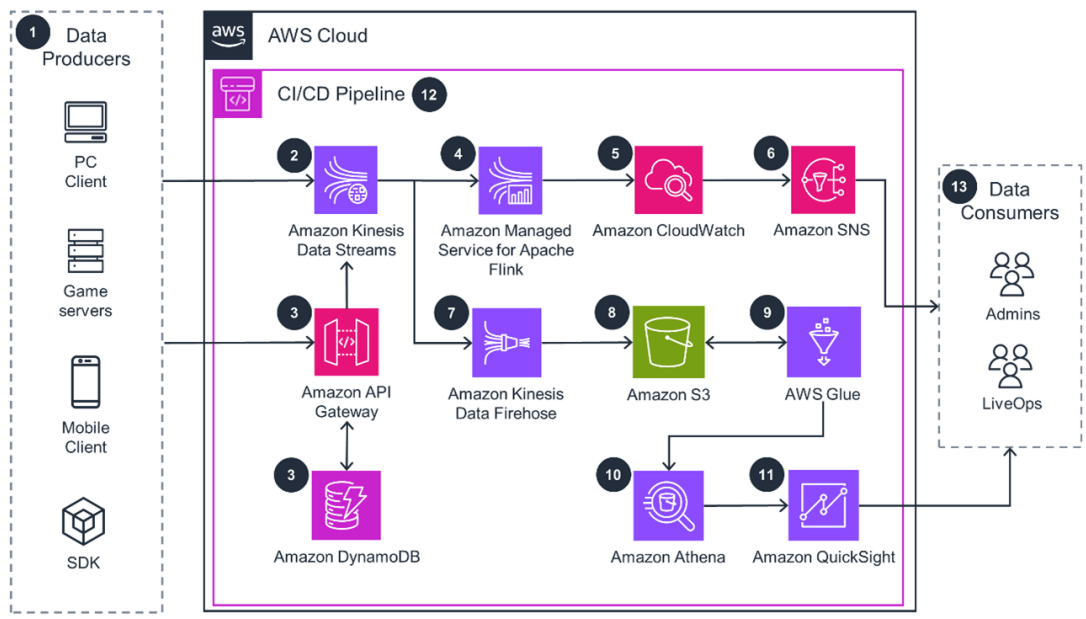
The Guidance for Game Analytics Pipeline on AWS helped Notorious Studios align with AWS best practices and take a modular approach. The team used [AWS Cloud Development Kit](#) (AWS CDK) to define cloud application resources using familiar programming languages (see figure 1. Notorious Studios’ game analytics pipeline).

For cleaning and processing data, Notorious Studios uses [AWS Glue](#), helping the company discover, prepare, and integrate all its data at any scale. To translate its data into Apache Parquet, Notorious Studios adopted [Amazon Kinesis Firehose](#), which reliably loads near real-time streams into data lakes, warehouses, and analytics services. The team can then extract analytics using [Amazon Kinesis Data Analytics](#) to process and analyze streaming data using standard SQL.

Notorious Studios also created data overlays using [Amazon QuickSight](#), which powers data-driven organizations with unified business intelligence at hyperscale. Using these solutions, Notorious Studios customized a game analytics pipeline in 2 days and deployed it in under 10 minutes. “The top benefit is how simple it was to get our pipeline running,” says Perez. “We can add and visualize data with ease by using Amazon QuickSight.”

Architecture Diagram





Outcome | Balancing Game Design through Near Real-Time Insights

Using analytics from its pipeline, Notorious Studios has found trends in player engagement, such as popular classes and map regions. The company uses these insights to balance the game design and deliver features that players will enjoy. “When our designers have a new question, we can query past playtests and answer them with real data,” says Perez.

Learn more

To learn more, visit <https://aws.amazon.com/solutions/guidance/game-analytics-pipeline-on-aws/>.

About Notorious Studios

Video game studio Notorious Studios is led by a group of experienced developers who work backward to deliver great gameplay and unite players within deep worlds.

AWS Services Used

Game Analytics Pipeline on AWS

This Guidance helps game developers implement a codified, modular, and serverless analytics pipeline that ingests telemetry events from game clients and backend services.



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